Fear in the Headlines: Measuring News Fear on a Reader's Own Scale, 2003–2026

Rahheb Shaikh · Updated 3 November 2026 · Final report, project ongoing

Abstract

Did the news become more frightening after COVID-19, and if so, why? We scored 3.35 million headlines from the New York Times (2003–2026), the Guardian (2003–2026) and ABC News Australia (2003–2021) for fear on a 0–3 scale defined by one reader's explicit codebook. A compact sentence model, distilled from a larger teacher trained on about 3,900 labelled headlines, reproduced codebook labels on unseen, scary-enriched sealed test sets with quadratic-weighted κ of .93 (2003–21 news) and .88 (2022–26 news), against κ .93 between two separate AI labelling passes. The share of scary headlines (level 2 or 3) in NYT news rose from 7.1% in 2017–19 to 10.1% in 2024–26 (+3.0 points, 95% CI 2.7 to 3.3); the Guardian rose by the same amount (+3.0, CI 1.9 to 3.9). COVID produced a temporary spike in frightening health coverage that had faded by 2024. A shift-share decomposition attributes about 90% of the lasting NYT rise to agenda (more coverage of war, terrorism and disasters) rather than to framing (how frightening each subject is made to sound). War and disaster coverage rose from about 1 in 8 NYT news headlines to about 1 in 5. Change-point analysis, which is exploratory, dates the last sustained step to July 2019, before COVID, and shows the scary share holding near 10% since February 2024, its highest sustained level in the 24-year series. Opinion pages showed a COVID bump but no lasting rise. We release the method and codebook so the measure can be reused.

1 Introduction

Headlines are the part of the news most people actually read, and a frightening headline shapes perceived risk even when the article beneath it is measured. Whether news has become more frightening matters for public anxiety, trust in media and how societies weigh real risks.

1.1 The problem with existing measures

Most large-scale studies measure sentiment or general negativity with lexicons or classifiers trained on generic emotion labels. Those tools confuse figurative harm ("a crushing defeat") with real threat, treat a court verdict over an old killing like the killing itself, and cannot say whose sense of fear they encode. We instead measure fear on an explicit, written scale anchored to one reader's judgement, and test the model against that scale on data it has never seen.

1.2 Research questions

1. **RQ1.** Did the share of frightening headlines change before, during and after COVID-19 (2017–2026)?
2. **RQ2.** Which subjects carry the frightening headlines, and did their share of coverage change?
3. **RQ3.** Is any change driven by agenda (which subjects are covered) or by framing (how frightening a given subject is made to sound)?

1.3 Contributions

- A written four-level fear codebook with tie-break rules, and a labelled corpus of about 3,900 headlines on that scale.
- An offline, phone-sized fear model (MiniLM, about 90 MB) that matches codebook labels closely on sealed tests, plus a 10-subject classifier sharing the same encoder.
- A scored analysis of 3.35 million headlines across three outlets and 24 years, and a pre-registered-style analysis of the COVID era with day-clustered uncertainty and a shift-share decomposition.

2 Background

2.1 Negativity and fear in the news

Audiences attend more to negative news than to positive news (Soroka & McAdams, 2015; Trussler & Soroka, 2014), and negative words raise click-through on headlines (Robertson et al., 2023). Cultivation research argues that heavy exposure to violent media fosters a "mean world" view (Gerbner & Gross, 1976). Long-run studies of headline sentiment report rising negativity and fear-related emotion in US outlets from 2000 to 2019 (Rozado, Hughes & Halberstadt, 2022). These studies use generic sentiment or emotion labels; ours uses a fear-specific scale tied to real threat.

2.2 Agenda versus framing

Agenda-setting holds that the media shape what audiences think about by choosing what to cover (McCombs & Shaw, 1972). Framing concerns how a covered issue is presented (Entman, 1993). A rise in frightening headlines could come from either: more coverage of threatening subjects, or more alarming treatment of the same subjects. A shift-share decomposition separates the two.

2.3 Measuring emotion in text

Lexicon methods are cheap but blind to context, such as "shooting" in basketball, or a verdict long after a crime. Sentence encoders (Reimers & Gurevych, 2019) capture context; large encoders such as MPNet (Song et al., 2020) are accurate but heavy, and knowledge distillation (Hinton, Vinyals & Dean, 2015) transfers their judgements into small models such as MiniLM (Wang et al., 2020). Ordinal scales are best trained with ordinal-aware losses (Cao, Mirjalili & Raschka, 2020) and evaluated with weighted kappa (Cohen, 1968).

3 Data

3.1 Sources

Source	Years	Headlines after deduplication	Access	Design
New York Times	Jan 2003 – Oct 2026	1,818,830	NYT Archive API	Full archive; news, opinion and other material types
ABC News (Australia)	Feb 2003 – Dec 2021	1,244,182	Public "A Million News Headlines" dataset	Full set; ends 2021
The Guardian	Jan 2003 – Oct 2026	289,833	Guardian Content API	One random day per month (all topics) plus complete archives for 10 threat-related tags
HuffPost News Category	2012 – 2022	210,000	Public dataset, CC BY 4.0 (Misra, 2022)	Used only to train the subject classifier

3.2 Collection

All data came through official APIs or openly licensed datasets, within published rate limits; no social-media platform was scraped. API responses were cached so every collection run is reproducible. The 2022–2026 extension was collected on 27 October 2026.

3.3 Deduplication

NYT and Guardian articles were deduplicated by article ID; Guardian random-day items without IDs, and ABC items, by date plus headline. A Guardian article reached by both the random-day and the tag designs is kept once, with both designs recorded. Headlines repeated five or more times within a source (for example "Corrections" or column names) are flagged as generic and excluded from the main analysis. Any headline used in training or in a test set was excluded from every other split.

3.4 Sampling designs and gaps

- Only the Guardian random-day sample is used for shares, because the tag archives deliberately over-sample threat topics.
- ABC ends in 2021, so post-COVID claims rest on the NYT and the Guardian.
- The NYT Archive API returned only about 700 articles for January 2026, against about 4,000 in a typical month, and 2026 runs to October.
- In 2025 the NYT changed its document-type label from "article" to "Article"; all filters are case-insensitive.

4 Measuring fear

4.1 The fear scale (codebook U1)

The question for every headline is: reading only this headline, how threatened or anxious would the reader feel? Levels are integers 0–3; when unsure between two levels, the lower is chosen.

Level	Meaning	Typical cases	Example
0 No fear	No threat to people	Sport, arts, procedure; rescued or found safe; figurative harm; animals only; natural-death obituaries	"Passengers escape plane inferno"
1 Mild concern	Harm that is contained, distant or past	Court cases, charges and verdicts over killings; inquiries; aftermath and costs; drought	"Man who killed two women jailed"
2 Clear fear	A real, current threat	A single accidental death; a missing person with the search ongoing; injuries; a fugitive on the run; large job losses	"River search fails to find missing driver"
3 Alarm	Death that has happened, or mass danger	Several deaths; any violent death or shooting; mass-casualty disasters	"Car bomb kills 5 in Iraqi market"

Headlines that sit between 2 and 3 (a search called off with people presumed dead, an explainer about a shooting) are scored 3 when labelled as integers; in analysis a model score of 2.5 or more counts as 3. Appendix A lists every rule.

4.2 Labelling and reliability

1. **Anchor ratings.** The reader rated 100 headlines by hand. A context check proposed 21 corrections where the reader lacked context (for example, a well-known disaster's death toll); the reader accepted all 21. The codebook was written from these ratings.
2. **Scale gap.** Generic AI raters scored about 0.44 levels higher than the reader on the same headlines (their 3 matched the reader's 2.2). This is why the model is trained on the reader's scale rather than on generic emotion labels.
3. **Corpus labels.** Further headlines were labelled with the codebook by an AI labeller (Claude), blind to all model scores, with doubtful cases collected and settled by the reader's rules. The training corpus holds about 3,900 headlines (3,118 train, 782 development, stratified by source and level).
4. **Reliability.** A separate AI labelling session, with the same codebook and no memory of the first labels, re-rated 150 headlines: exact agreement 86%, quadratic-weighted κ .93. This shows the rules give consistent answers. It isn't human agreement, and both passes may share blind spots.
5. **Check against the reader.** On the 20 of the reader's own hand ratings held out of training, the model scored κ .81 and 70% exact, and caught 4 of 7 scary items. The sample is small.

4.3 Model

The production model is small enough to run offline on a phone; a larger model supervises it during training.

1. **Teacher.** A sentence encoder (all-mpnet-base-v2; Song et al., 2020) fine-tuned with an ordinal (CORAL) head, a pairwise similarity loss and synthetic minimal pairs, which are edited headlines that change one factor, such as a death becoming a rescue. It is trained on the reader-scale labels (weight 2) plus earlier generic-rater labels (weight 0.5). Five seeds are averaged.
2. **Student.** MiniLM-L6 (Wang et al., 2020; inputs up to 32 tokens) distilled by mean-squared error on the teacher's scores for about 300,000 unlabelled ABC headlines, with the reader-scale labels added as direct targets (weight 3). A story-type head is trained jointly, and the final fear head is a ridge regression on the student's embeddings.

3. **Cut-points.** The continuous score is mapped to levels by three cut-points. The 1/2 cut maximises recall of scary headlines subject to at most 15% false alarms on "tricky calm" headlines (calm headlines containing harm words); the other two cuts maximise weighted κ. Cuts are tuned on the development set with 5-fold cross-validation; the final cuts are 0.625, 1.225 and 2.475, remapped to the level boundaries.

4.4 Subject classifier

Each headline also gets one of 10 subjects: war/terror, disaster/accident, crime/justice, health, economy, politics, environment, science/tech, sport, culture/lifestyle. Training labels come from the outlets' own metadata: NYT sections and subject keywords, Guardian sections and tags, and HuffPost categories, mapped by fixed rules (323,000 balanced headlines). A logistic-regression head on the same student encoder predicts the subject, so fear and subject come from one pass. On 193 hand-labelled headlines it is right 67% of the time (ABC 71%), and 85% when its confidence is at least 0.6, which holds for about half of headlines. The outlets' own tags agree with the hand labels 73% of the time.

5 Model evaluation

5.1 Protocol

- **Sealed test sets.** Each set holds 300 NYT headlines that no model had seen: 150 scary (level 2–3) and 150 calm, half of the calm ones "tricky" (they contain harm words). Candidates were drawn without any model, using a keyword pool plus a random draw. They were labelled blind on the codebook, locked, and opened once. Five sets were used over the project; sets #4 (2003–21) and #5 (2022–26) test the final model.
- **Frozen plans.** Every training change, gate and comparison was written down before its numbers were computed. A model replaced its predecessor only if it passed the pre-set gates on a sealed set.
- **Enrichment.** The sets are half scary, while ordinary news is about 10% scary, so these figures don't describe everyday accuracy. Section 5.4 does.
- **Metrics.** Quadratic-weighted κ against the labels; exact level accuracy, overall and per level; recall and precision of scary headlines; the false-alarm rate on tricky calm headlines; AUC for scary versus calm.

5.2 Results

Metric	Sealed #4 (NYT 2003–21)	Sealed #5 (NYT 2022–26)
Weighted κ	.93	.88
Exact level	81%	77%
True 0 / 1 / 2 / 3 right	– / 48% / 77% / 83%	95% / 55% / 69% / 78%
Scary headlines caught (recall)	94%	87%
Precision of "scary"	92%	87%
False alarms on tricky calm	17%	23%
AUC, scary vs calm	.98	.95

The final model (v7e) cuts the exact-level error from 60% to 19% compared with the earlier grammar-and-encoder model (v6), which scored κ .60, 40% exact and 41% false alarms on sealed #4. Appendix B lists every step in between.

5.3 Errors

- **Mild news is the weak level.** About a third of true level-1 headlines (court cases, aftermath, distant tension) are rated 2. In a 600-headline sample of 2022–26 headlines picked near the model's 1/2 boundary, 154 of the 337 it rated 2 were true 1s.
- **Newer news is harder.** Agreement drops from κ .93 to .88 on 2022–26 headlines, plausibly because of new topics (the wars in Ukraine and Gaza, the 2026 Iran war) and newer headline styles.
- **Two fixes were tried and rejected on sealed #5.** Retraining on 197,000 recent headlines plus 600 new labels (v7f) caught more scary news (91%) but got fewer 1s right (38%). Raising the 1/2 cut-point got more 1s right (66%) but missed more scary news (81%). Neither passed the pre-set bar, so v7e was kept.

5.4 Accuracy on the ordinary news mix

Sealed items were drawn from keyword-matched and non-keyword headlines, so reweighting them by each group's real share of NYT headlines estimates ordinary-news performance. An earlier version of this report treated the non-keyword items alone as a random sample, which was wrong and has been corrected here.

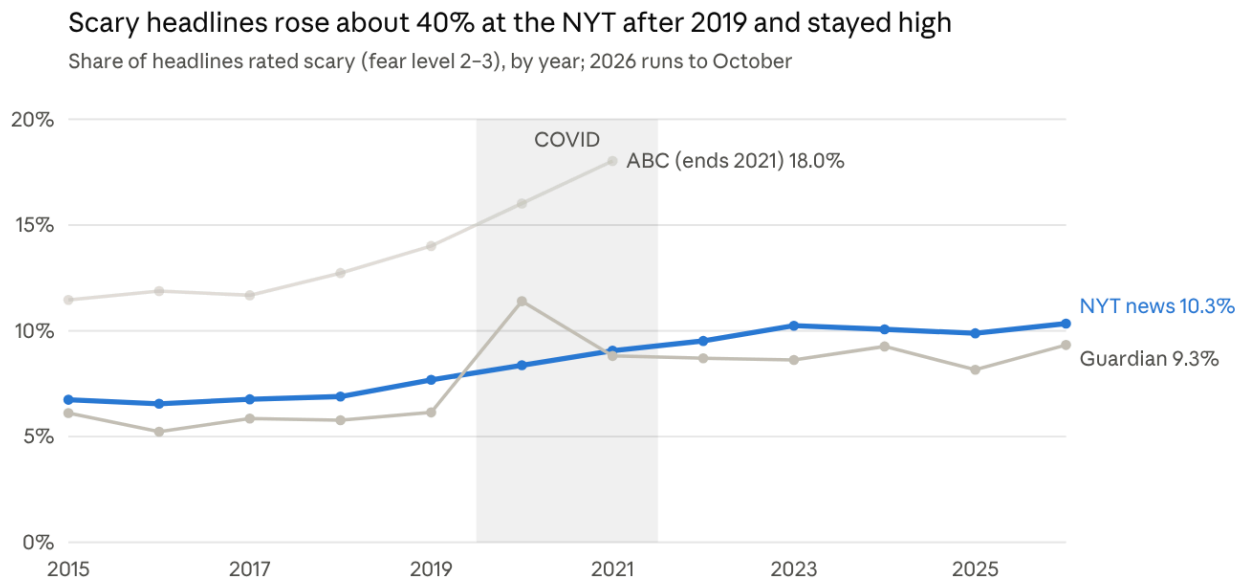
Pool	True scary share	Model's scary share	Weighted recall	Weighted precision
Sealed #4 (2003–21)	3.0%	5.5%	96%	51%
Sealed #5 (2022–26)	9.3% (95% CI 4.9 to 14.5)	10.2%	60%	55%

The model's overall share is close to the labels for recent news and too high for older news. So overall shares are usable, but on ordinary news only about half of individually flagged headlines are truly scary. These estimates are rough, because few non-keyword items were scary (5 in sealed #5).

6 Results

All analyses follow a plan frozen before any number was computed. Periods are fixed: pre-COVID 2017–19, COVID 2020–21, after 2022–23, recent 2024–26. Intervals are 95% day-clustered bootstrap intervals (1,000 resamples).

6.1 Trends over time (RQ1)



At the NYT the rise begins in 2019 and continues steadily through COVID, then levels off at about 10% from 2023. The Guardian jumps in 2020 and settles about 3 points above its pre-COVID level. ABC, whose data end in 2021, rises from 2017 onward. Each outlet has its own baseline, so changes, not levels, are compared.

A daily view shows why this can feel constant. In 2024–26, NYT news carried at least one scary headline on 100% of the 984 days with data, averaging 9.8 a day (median 9), up from 7.4 a day in 2017–19 (99.8% of days). About 90% of its news headlines were still not scary. This daily view was computed after the frozen plan and is descriptive only.

6.2 Before, during and after COVID

Sample	Pre-COVID 2017–19	COVID 2020–21	After 2022–23	Recent 2024–26	Recent minus pre (95% CI)
NYT news, scary (2–3)	7.1%	8.7%	9.9%	10.1%	+3.0 (2.7 to 3.3)
NYT news, alarm (3)	1.6%	1.5%	2.0%	2.1%	+0.5 (0.4 to 0.6)
Guardian, scary (2–3)	5.9%	10.3%	8.7%	8.9%	+3.0 (1.9 to 3.9)
Guardian, alarm (3)	1.0%	1.3%	1.4%	1.5%	+0.5 (0.2 to 0.9)
NYT opinion, scary (2–3)	2.1%	2.9%	2.5%	1.9%	–0.2 (–0.6 to 0.1)

News became scarier and stayed scarier at both outlets. Opinion headlines rose during COVID (+0.8 points, CI 0.4 to 1.2) and then returned to baseline.

6.3 Subjects (RQ2)

Subject, NYT news	Share of headlines 2017–19	2020–21	2022–23	2024–26	Scary within subject 2017–19	2024–26
War / terror	7.6%	6.1%	12.0%	12.0%	31%	32%
Disaster / accident	4.3%	4.7%	5.9%	7.0%	45%	49%
Crime / justice	8.1%	7.3%	8.6%	8.7%	26%	23%
Health	6.0%	16.0%	7.4%	5.7%	6%	8%
Politics	15.1%	14.3%	12.2%	17.9%	1%	1%
Economy	9.0%	8.3%	9.1%	8.6%	1%	1%
Culture / lifestyle	26.5%	21.4%	22.4%	22.6%	0%	0%
Sport	11.3%	9.7%	9.6%	4.8%	0%	0%

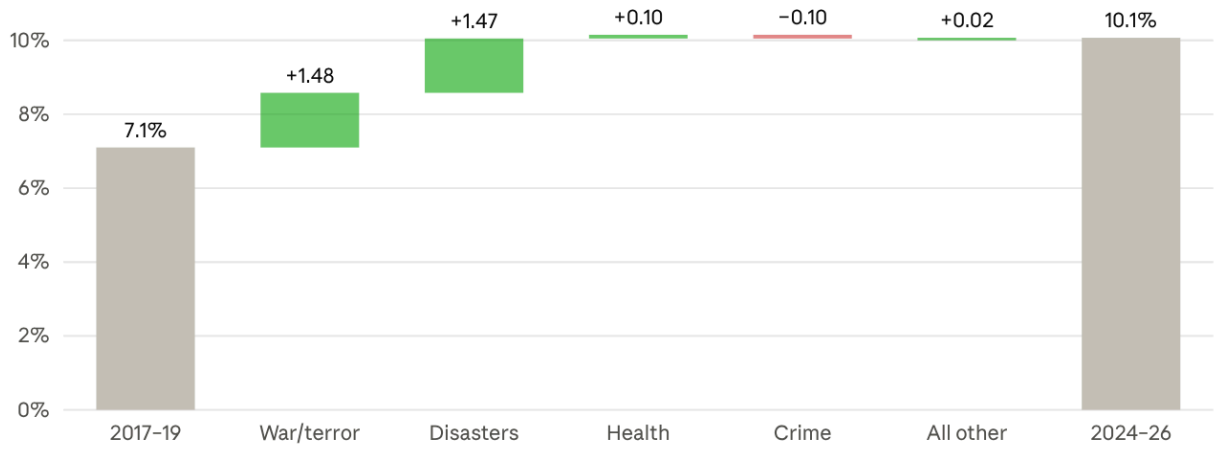
Four subjects carry almost all frightening headlines: disaster/accident (about half of its headlines are scary), war/terror and crime/justice (about a quarter to a third each), and health during COVID (15% scary in 2020–21). War/terror coverage rose from 7.6% to 12.0% of NYT news after 2022, and disaster coverage from 4.3% to 7.0%. Health returned to its pre-COVID share and tone by 2024–26. The Guardian shows the same pattern: war/terror rose from 5.6% to 9.9% of headlines and disasters from 3.0% to 3.8%.

6.4 Agenda or framing? (RQ3)

Figure 2. War and disasters account for almost all of the 3-point rise

NYT news, share of headlines rated scary; each step = the subject's change in scary headlines, in points

Green adds to the scary share, red takes away; grey bars are the period totals



A shift-share decomposition splits the change in the scary share into agenda (subject shares change, scariness held at pre-COVID levels), framing (scariness within subjects changes, shares held) and their interaction. For NYT news the +2.97-point rise splits into agenda +2.73, framing +0.13 and interaction +0.11, so about 92% is agenda. For the Guardian the +2.95-point rise splits into agenda +1.96, framing +0.77 and interaction +0.22, so about two-thirds is agenda. For NYT opinion the total change is –0.22 points, with no meaningful component.

6.5 Long view (ABC, 2003–2021)

ABC's scary share was 12–14% from 2003 to 2009, fell to about 10.5% in 2012–14, and then climbed to 14.0% in 2019, 16.0% in 2020 and 18.0% in 2021. The post-2014 rise at ABC began well before COVID, consistent with the NYT's pre-pandemic upturn.

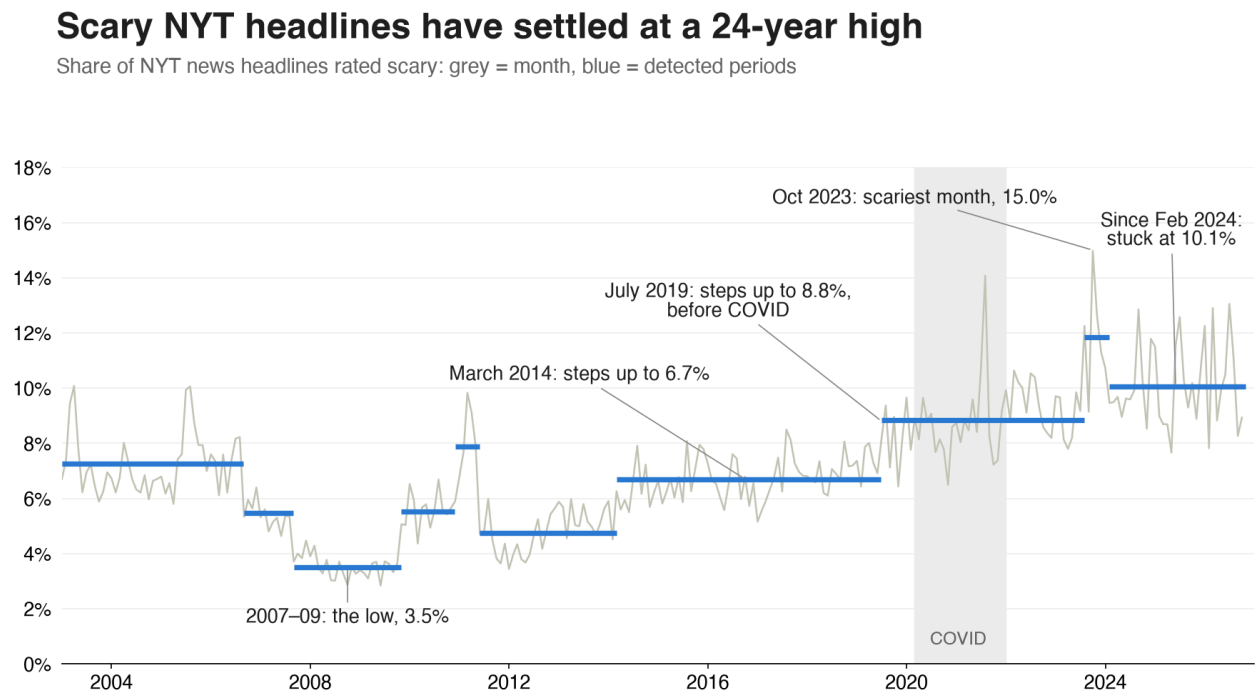
6.6 Robustness

Check	NYT news, pre → recent	Guardian, pre → recent
Main analysis (scary share)	7.1% → 10.1%	5.9% → 8.9%
Mean fear score instead of share	0.32 → 0.42	0.31 → 0.41
Including generic repeated headlines	6.9% → 9.8%	5.9% → 8.8%
Confident subjects only (conf ≥ 0.6)	11.6% → 17.8%	7.6% → 12.0%
Outlet's own tags as subject: total change, agenda part	+3.4 points; agenda +4.8	+5.4 points; agenda +1.0*

Every check points the same way. *Guardian tags are complete only for the threat-tag design, so this check is weaker there; its large interaction term (+4.1) reflects that design.

6.7 Key transitions (exploratory)

This analysis was planned after the main results were seen, so it is exploratory. A change-point search on monthly NYT news shares found these periods:



Period	Scary share	What marks it
Sep 2007 – Oct 2009	3.5%	The low. Economy news is 20% of headlines and war only 5.5%
Dec 2010 – May 2011	7.9%	A brief spike that fell back to 4.8% within months
Mar 2014 – Jun 2019	6.7%	A step up, with disaster headlines getting scarier (29% to 42% scary)
Jul 2019 – Jul 2023	8.8%	A step up about eight months before COVID, including its health surge
Aug 2023 – Jan 2024	11.8%	The peak: Maui and Gaza. October 2023 is the scariest month at 15.0%
Feb 2024 – Oct 2026	10.1%	The plateau, now 33 months long

The biggest change in coverage is war and disaster together: 11.9% of NYT news headlines in 2017–19 and 19.0% in 2024–26, or 12 a day rising to 18. By the outlets' own tags the rise is steeper, from 8.8% to 20.5%. These two subjects produced 72% of scary headlines in 2024–26, up from 60%.

6.8 Checks on the rise (exploratory)

- **Is it just fewer calm headlines?** Mostly not since 2017. Scary headlines rose from 7.4 to 9.8 a day (+33%) while calm ones fell about 8%, so about three-quarters of the rise is more scary headlines. Without sport, whose coverage shrank when the NYT closed its sports desk, the share still rises from 8.0% to 10.6%.
- **Over the long run, partly yes.** From 2007–09 to 2024–26, calm headlines halved (188 to 86 a day) while scary ones rose about 20%, so most of that rise reflects fewer calm headlines. Earlier archive years may also count different items, so these long comparisons are less reliable.
- **Does it depend on the model?** Blind labels alone also show a rise, from 4.2% (2017–19, CI 2.5 to 7.3) to 11.0% (2024–26, CI 5.5 to 18.1): +6.7 points, CI 0.3 to 14.4. The direction agrees with the model, but the size is very uncertain. Labelling rules were refined between pools, which could affect this comparison.
- **Is tone really flat?** Since 2017, yes: framing adds +0.13 of the 2.97-point rise. Over longer spans it matters more. From 2003–06 to 2024–26, framing explains about 44% of the rise, and disaster headlines went from 36% to 49% scary.
- **Could the shared encoder bias subjects?** Partly. The fear and subject heads use one encoder, and model subjects match outlet tags more often for scary headlines. But the agenda result is stronger, not weaker, when using the tags alone.

7 Discussion

7.1 What changed, and what did not

The share of scary headlines rose by about 3 points (a 40% relative rise at the NYT) and has stayed up since the pandemic ended. Since 2017 the tone of coverage within subjects has barely moved: a war or disaster headline was about as frightening in 2024–26 as in 2017–19. What changed is the mix, with war and disaster rising from about 1 in 8 NYT news headlines to about 1 in 5. Over longer spans tone matters more, especially for disasters (section 6.8).

These data measure coverage, not events. They can't say whether the world became more dangerous, whether newsrooms gave danger more attention, or both. That is the most important open question this work leaves.

Earlier spikes faded: the 2011 spike fell back within months. The current plateau has lasted 33 months, which suggests a new baseline rather than a passing event, although a new shock or a run of quieter events could still move it.

7.2 COVID as a spike, not a shift

COVID-19 raised fear through one subject, health, which both took more space and became more alarming. Both effects reversed by 2024. The last sustained step came in July 2019, before the pandemic, and the lasting rise is carried by other subjects. Accounts that blame a post-COVID "anxiety culture" in the press are not supported by these headlines.

7.3 News versus opinion

Opinion headlines rose only during COVID and then returned to their pre-COVID level. The lasting rise is in reporting of events, which fits an agenda explanation better than a deliberate change in editorial voice.

7.4 Measuring on a reader's scale

A scale written from one reader's judgement makes the measure explicit and testable: anyone can read the codebook, disagree with a rule, and re-label. Generic raters scored about half a level more fearful than our reader, which would inflate absolute levels but, because the bias is roughly constant, should leave trends similar. Agreement on sealed sets suggests a compact, offline model can apply a written scale consistently in aggregate, although individual flags on ordinary news are much less reliable (section 5.4).

8 Limitations

- **One reader's scale.** Levels reflect one person's codebook. Other readers may draw the lines elsewhere; the codebook makes the choices visible but does not make them universal.
- **AI-assisted labels.** Most corpus labels were produced by an AI labeller following the codebook. The κ .93 consistency check compared two AI sessions, not people, and direct comparison with the reader's own ratings is limited to 20 held-out items.
- **Headlines only.** Article text, images, placement and reach are not measured, so the results describe what headlines signal, not what readers saw most.
- **Coverage after 2021.** Only the NYT and the Guardian extend past 2021, and both are English-language, centre-left outlets. Other outlets, and television or social platforms, may differ.
- **Model error over time.** Accuracy is lower on 2022–26 news (κ .88). Trends assume errors are stable across years. The check in section 5.4 suggests the model overstates older news more than recent news, which would make the true rise larger, but those estimates are rough.
- **Coverage is not reality.** These data can't separate changes in world events from changes in newsroom attention.
- **Individual headlines.** On ordinary news, only about half of headlines the model flags as scary are truly scary. Results are meant for aggregate use.
- **Subject error.** About one subject in three is wrong. The main conclusion holds with the outlets' own tags, but finer subject comparisons should be read with care.
- **Collection artefacts.** January 2026 is under-collected at the NYT, 2026 is partial, and the NYT's 2023 closure of its sports desk shrinks sport's share for reasons unrelated to fear.

9 Ethics, data and code availability

- **Data.** Only published headlines and outlet metadata were used, collected through official APIs within their terms or from openly licensed datasets. No personal data and no social-media posts were collected.
- **Redistribution.** NYT and Guardian headline text is subject to their API terms; this repository shares code, the codebook and aggregate results, and NYT and Guardian headlines can be re-fetched from the same APIs. Other cleaned data, scores and labels are available on request (see `DATA_ACCESS.md`). ABC and HuffPost datasets are public.
- **Labelling.** No crowd workers were used. The reader's own ratings anchor the scale; AI-assisted labels are disclosed above.
- **Intended use.** The measure describes coverage. It should not be used to rank or penalise individual journalists or articles.

10 Conclusion

Across 3.35 million headlines, scary news rose by about 3 points of all headlines after 2019 at both the NYT and the Guardian, and it has not come down. COVID-19 drove a temporary spike in frightening health news, but the lasting rise comes from what is covered (more war and disaster) rather than from how it is told. War and disaster went from about 1 in 8 NYT news headlines to about 1 in 5, and the scary share has held near 10% for 33 months, its highest sustained level in 24 years of data. Whether that reflects a more dangerous world or more attention to danger is the next question. A compact offline model trained on an explicit fear scale can track this in aggregate, which could let readers monitor the fear in their own news diet.

References

- Cao, W., Mirjalili, V., & Raschka, S. (2020). Rank consistent ordinal regression for neural networks with application to age estimation. *Pattern Recognition Letters, 140*, 325–331.
- Cohen, J. (1968). Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit. *Psychological Bulletin, 70*(4), 213–220.
- Entman, R. M. (1993). Framing: Toward clarification of a fractured paradigm. *Journal of Communication, 43*(4), 51–58.
- Gerbner, G., & Gross, L. (1976). Living with television: The violence profile. *Journal of Communication, 26*(2), 172–199.
- Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network. arXiv:1503.02531.
- Kulkarni, R. (2018). *A Million News Headlines* (Version 6.0) [Data set]. Harvard Dataverse. https://doi.org/10.7910/DVN/SYBGZL
- McCombs, M. E., & Shaw, D. L. (1972). The agenda-setting function of mass media. *Public Opinion Quarterly, 36*(2), 176–187.
- Misra, R. (2022). News Category Dataset. arXiv:2209.11429.
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *Proceedings of EMNLP-IJCNLP 2019*, 3980–3990.
- Robertson, C. E., Pröllochs, N., Schwarzenegger, K., Pärnamets, P., Van Bavel, J. J., & Feuerriegel, S. (2023). Negativity drives online news consumption. *Nature Human Behaviour,*

7(5), 812–822.
- Rozado, D., Hughes, R., & Halberstadt, J. (2022). Longitudinal analysis of sentiment and emotion in news media headlines using automated labelling with Transformer language models. *PLOS ONE, 17*(10), e0276367.
- Song, K., Tan, X., Qin, T., Lu, J., & Liu, T.-Y. (2020). MPNet: Masked and permuted pre-training for language understanding. *Advances in Neural Information Processing Systems 33*.
- Soroka, S., & McAdams, S. (2015). News, politics, and negativity. *Political Communication, 32*(1), 1–22.
- Trussler, M., & Soroka, S. (2014). Consumer demand for cynical and negative news frames. *The International Journal of Press/Politics, 19*(3), 360–379.
- Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., & Zhou, M. (2020). MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained Transformers. *Advances in Neural Information Processing Systems 33*.

Data services: The New York Times Archive API; The Guardian Open Platform (Content API). Reference details were checked against Crossref, arXiv and DataCite on 3 November 2026.

Appendix A: Codebook rules in full

- **0:** no threat to people; sport, arts, procedure; rescued, found safe or escaped; figurative harm words ("fights for life after poll thrashing"); animals only; history pieces; obituaries for old age or natural death; a fugitive who has been caught.
- **1:** an investigation, inquiry, hearing or inquest after harm; aftermath and costs; a contained fire; drought; tension or threats far away; crime with a victim but no ongoing danger; charges, trials, verdicts or executions over killings, even mass or old ones.
- **2:** a single accidental death, a child included; a missing person with the search ongoing; a bombing with likely casualties; injuries; a child's suspicious death; hundreds of local jobs lost or large layoffs; an ongoing local emergency; a fugitive still on the run; an aftermath piece about a known attack that names no death.
- **3:** several deaths, accidental or not, stated as fact; any violent death, shooting or critical wounding by violence, including of militants or suspects and of a single civilian far away; many people missing with live danger; mass-casualty disasters.
- **Tie-breaks:** use context the reader would have (a well-known disaster carries its toll); if the threat is over, score the remaining worry; if torn between two levels, choose the lower; near-death cases that are not a stated death (search called off, presumed dead) are 2.5–2.8 and count as 3.

Appendix B: Model development history

Model	Change	Key test result	Decision
v6	Grammar rules + MiniLM, trained on disaster labels	Sealed #4: κ .60, 40% exact	Earlier baseline
v10 (teacher)	MPNet with ordinal head and synthetic minimal pairs	Sealed #1: AUC .97, 65% of 3s caught	Became the teacher
v7	MiniLM student, smooth calibration to the reader	Sealed #1: only 28% of 3s caught (calibration flattened the top)	Replaced
v7b	Level cut-points instead of calibration	Sealed #3: κ .83, 75% of 2–3s caught	Made default
v7c	+ 600 labels near the 1/2 boundary	Sealed #4: κ .85, 53% of true 2s right	Made default
v7d	+ 300 accident and rescue labels	Development κ .81 (gate failed)	Rejected
v7e	Teacher retrained directly on the reader's scale	Sealed #4: κ .93; sealed #5: κ .88	**Final model**
v7f	+ 197,000 recent headlines and 600 new labels	Sealed #5: κ .88, but only 38% of 1s right	Rejected

Appendix C: Reproducibility

1. **Collection:** `outlets_fetch.py`, `guardian_targeted.py`, `recent_fetch.py` (APIs, cached and resumable).
2. **Labels and tests:** `prep_scary_test*.py` (sealed pools), `prep_v7f_data.py`, `build_u1_corpus.py`.
3. **Models:** `train_v12.py` (teacher), `build_fear_v7e.py` (student), `build_subject.py` and `build_subject_rules.py` (subjects).
4. **Scoring and analysis:** `score_all_v7e.py` (3.35 million rows), `analysis_covid.py` (frozen plan, 1,000 day-cluster bootstrap resamples, seed 0), `analysis_transitions.py` and `analysis_transitions2.py` (exploratory).

The production model runs on CPU with no network access; scoring all 3.35 million headlines took 16 minutes on a laptop.